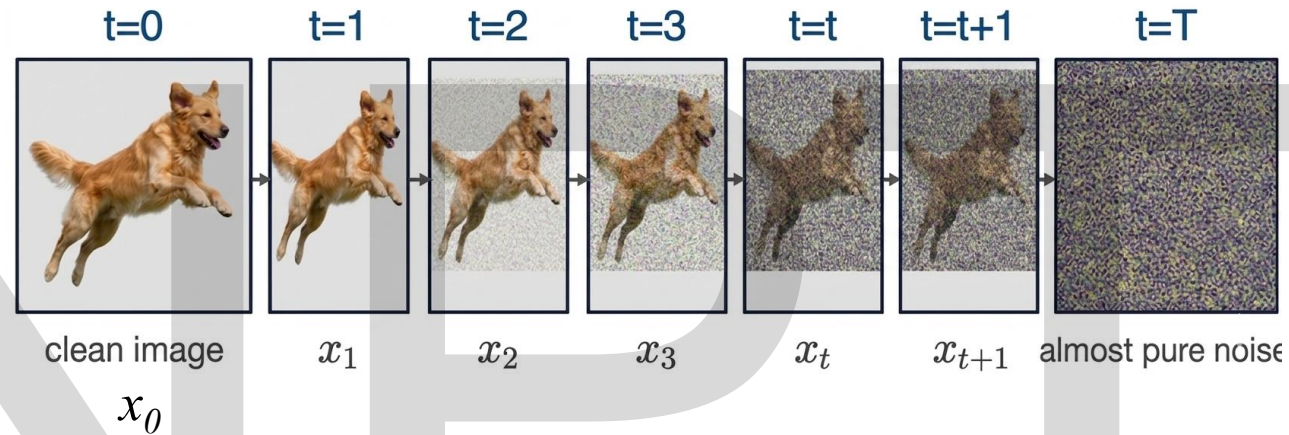


Mathematical Foundations of Diffusion Models

Setting up the Problem Statement

Forward Diffusion Process



$q(x_t|x_{t-1})$: forward diffusion process $\rightarrow$ fixed, known



The probability distribution of x_t conditioned on x_{t-1}

x_0 : Original Data(image or signal)

x_t : Noisy version at time step t

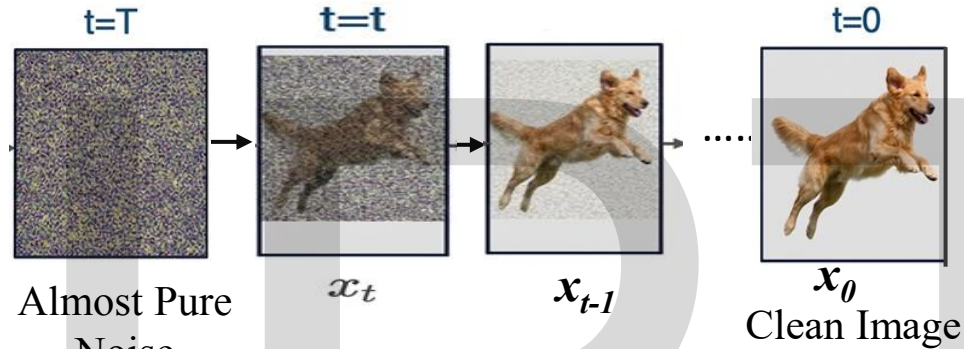
T : Total number of diffusion steps

$t \in \{1, 2, \dots, T\}$

- ✓ Forward diffusion: generate a noisier sample from a cleaner sample.
- ✓ Since the forward diffusion process is known and fixed, this distribution is predefined as Gaussian

Setting up the Problem Statement

Reverse Diffusion Process



$p_{\theta}(x_{t-1}|x_t)$: Reverse diffusion process $\rightarrow$ learning happens

The probability distribution of x_{t-1} conditioned on x_t , parameterized by θ

- ✓ θ refers to all the trainable weights of the denoising neural network
- ✓ Reverse Diffusion generates cleaner sample from a noisier sample

Noise Variance Schedule

- Noise variance schedule is the sequence of values $\{\beta_1, \beta_2, \dots, \beta_T\}$

- ✓ It controls how much noise is added at each diffusion step.

β_t : Noise variance **schedule** $\beta_t \in (0,1)$



*It is **not constant**—it changes with time t*

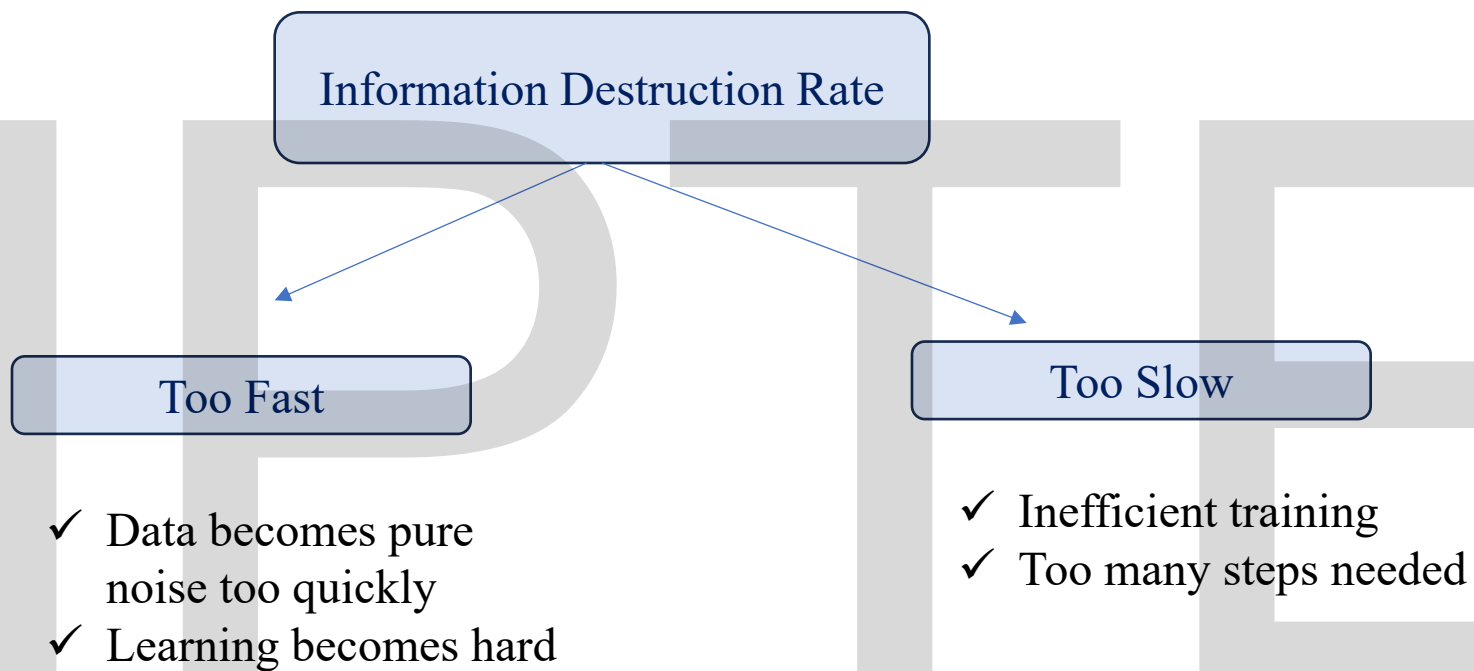
β controls the rate at which noise is injected and signal-to-noise ratio decreases.

Small $t \rightarrow$ small noise

Large $t \rightarrow$ larger noise

This gradual increase is the “schedule”

Why is the Schedule Important?



We want:

$$x_T \sim \mathcal{N}(0, I)$$

Schedule must ensure:

complete loss of structure by step T

Review of Gaussian Distribution

A **Gaussian distribution** is a probability distribution that describes data that clusters around a mean with symmetric spread.

x : random variable (data point)

μ : **mean**

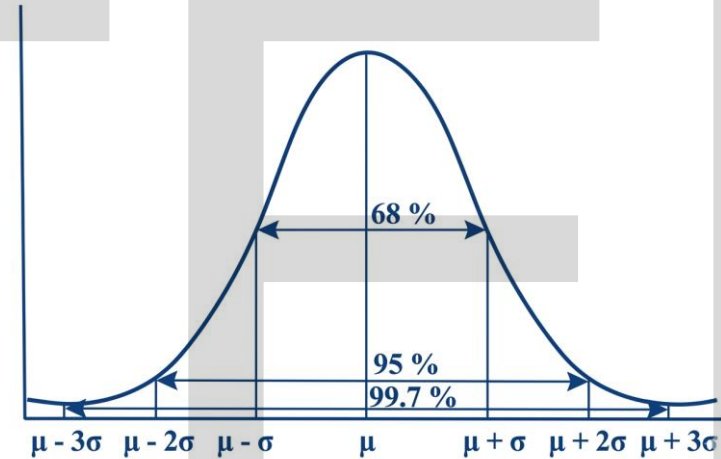
σ^2 : **variance** (spread of data)

σ : standard deviation

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Normalization
factor
Ensures the total
probability=1

Probability density decreases
exponentially as distance from the
mean increases.



Standard Gaussian

Then:

$$\mu = 0, \sigma^2 = 1$$

$$\sim \mathcal{N}(0, 1)$$

Mean

Variance

Multivariate Gaussian

A Multivariate Gaussian Distribution is a generalization of the Gaussian distribution to multiple variables (a vector).

Instead of a single number x , we now have:

$$x \in \mathbb{R}^d$$

$$p(x) = \mathcal{N}(x; \mu, \Sigma)$$



μ : Mean Vector

Σ : Covariance matrix

The probability density function modeled as a multivariate normal distribution with mean vector μ and covariance matrix Σ

Multivariate Gaussian

In Machine Learning, why do we need Multivariate Gaussian?

In ML, data is rarely a single variable.

- Image → millions of pixels
- Embeddings → high-dimensional vectors

$$X \in \mathbb{R}^d$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ \vdots \\ \vdots \\ x_d \end{bmatrix}$$

Real-world data has dependencies:

- ✓ Pixels in an image are correlated
- ✓ Features in datasets are related

Relationship between features is captured by Covariance

Covariance

Covariance measures how two variables change together.



ICE CREAM SALE IN HOT
SUMMER!!!
(positive Covariance)



HOT CHOCOLATE SALE IN HOT
SUMMER!!!
(Negative Covariance)

Covariance matrix

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Covar}(x_1, x_2) \\ \text{Covar}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

Covariance Matrix-Numerical Example

Example: Height and Weight Dataset

Suppose each datapoint is:

$$x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \text{height (cm)} \\ \text{weight (kg)} \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Covar}(x_1, x_2) \\ \text{Covar}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

Person	Height (x_1)	Weight (x_2)
A	150	50
B	160	55
C	170	65
D	180	72

Mean Height (μ_1)	Mean Weight (μ_2)
$(150+160+170+180)/4 = 165$	$(50+55+65+72)/4 = 60.5$

$$\text{Var}(x_1) = E[(x_1 - \mu_1)^2]$$

Person	X_1	$X_1 - \mu_1$	$(X_1 - \mu_1)^2$
A	150	-15	225
B	160	-5	25
C	170	5	25
D	180	15	225

$$\begin{aligned} \text{Var}(x_1) &= \frac{225+25+25+225}{4} \\ &= 125 \text{ (population variance)} \end{aligned}$$

$$\text{Cov}(x_1, x_2) = E[(x_1 - \mu_1)(x_2 - \mu_2)]$$

Person	X_1	$X_1 - \mu_1$	X_2	$X_2 - \mu_2$	$(X_1 - \mu_1)(X_2 - \mu_2)$
A	150	-15	50	-10.5	157.5
B	160	-5	55	-5.5	27.5
C	170	5	65	4.5	22.5
D	180	15	72	11.5	172.5

$$\mu_1 = 165$$

$$\mu_2 = 60.5$$

$$\text{Cov}(x_1, x_2) = E[(x_1 - \mu_1)(x_2 - \mu_2)] = (157.5 + 27.5 + 22.5 + 172.5) / 4 = 380 / 4 = 95$$

Covariance matrix

$$\Sigma = \begin{bmatrix} \text{Var}(x_1) & \text{Covar}(x_1, x_2) \\ \text{Covar}(x_2, x_1) & \text{Var}(x_2) \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 125 & 95 \\ 95 & 73.25 \end{bmatrix}$$

NPTEL

Summary

Next Session

Denoising Diffusion Probabilistic Model- Forward Process